Explicit asymptotically good relaxed locally correctable codes with near-log-squared query complexity

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Allowed standard prerequisites (editorial).

- Explicit lossless conductors / unique-neighbor expander construction. Author Theorem6 citing [8,Theorem7.3]; exact quantitative statement included.
- Standard RLCC error amplification. Author Claim4 explicitly cites [12] for its elementary proof; precise statement retained.

3 Author preliminaries

Unless stated otherwise, all logarithms in this paper are taken to the base 2. The set of natural numbers is $\mathbb{N} = \{0, 1, 2, \ldots\}$. For $n \in \mathbb{N}$, $n \geq 1$, we use $[n]$ to denote the set $\{1, \ldots, n\}$. For $q \in \mathbb{N}$, $q \geq 2$, we use H_q to denote the q-ary entropy function, and $H = H_2$ to denote the binary entropy function. For a finite set N, we refer to a function $v \in \mathbb{F}^N$ as a vector and we say that it is indexed by N. For a vector $v \in \mathbb{F}^N$ and $i \in N$ we use v_i as a shorthand for $v(i)$. For a vector $v \in \mathbb{F}^N$ and a set $N' \subseteq N$ we denote by $v_{N'}$ the vector $v' \in \mathbb{F}^{N'}$ such that $v'_i = v_i$ for every $i \in N'$. For two vectors $u, v \in \mathbb{F}^N$, their (absolute) hamming distance is $|\{i \in N \mid u_i \neq v_i\}|$, which we denote by $\mathrm{Dist}(u, v)$, and their relative hamming distance is $\frac{\mathrm{Dist}(u,v)}{|N|}$, which we denote by $\mathrm{RelDist}(u, v)$.

Definition 2. For a finite set N of size $|N| = n$ and a field $\mathbb{F}$, a code is a linear subspace $C \subseteq \mathbb{F}^N$. We say that the code C is indexed by N and that it is over $\mathbb{F}$. The length of the code is n. The dimension of the code, usually denoted by k, is the dimension of C over $\mathbb{F}$. The (non-local) distance of the code, denoted by d, is $\min_{c,c' \in C, c \neq c'} \mathrm{Dist}(c, c')$. The rate of the code, typically denoted by ρ, is $\frac{k}{n}$. The (non-local) relative-distance of the code is defined to be $\frac{d}{n}$. The elements of C are called codewords.

Definition 3. A code $C \subseteq \mathbb{F}^N$ is called a (q, δ, ε)-RLCC (relaxed locally correctable code, abbreviated) if there exists a randomized procedure $\mathrm{Cor} : \mathbb{F}^N \times N \to \mathbb{F} \cup \{\perp\}$ with the following guarantees:

- For every $i \in N$, $c \in C$ and $w \in \mathbb{F}^N$, satisfying $\mathrm{RelDist}(w, c) \leq \delta$, $\mathrm{Cor}(w, i) \in \{c_i, \perp\}$ with probability at least $1 - \varepsilon$.
- $\mathrm{Cor}(c, i) = c_i$ with probability one on any $c \in C$ and $i \in N$.
- $\mathrm{Cor}(w, i)$ always makes at most q queries to w.

We refer to Cor as the local corrector (or the corrector). The parameter δ is called the correction radius, and the parameter q is called the query complexity. The error parameter of an RLCC can be easily amplified at low cost to the query complexity, as stated in the following claim (for a simple proof see, e.g., [12]).

Claim 4. Let $C \subseteq \mathbb{F}^N$ be a (q, δ, ε)-RLCC. Then, for any $h \in \mathbb{N}$, C is also an $(hq, \delta, \varepsilon^h)$-RLCC. We set some standard notation. Let $G = (V, E)$ be an undirected graph. For $v \in V$ we define $\Gamma(v)$ as the set of neighbors of v in G, and let $\deg(v)$ be the degree of v. For a set of vertices $S \subseteq V$, we let $\Gamma(S) = \cup_{v \in S} \Gamma(v)$, and define

$$\Gamma_u(S) = \{v \in V \mid v \text{ is adjacent to exactly one } u \in S\}.$$

Definition 5 (Unique-neighbor expanders). A left-d-regular bipartite graph $G = (L, R, E)$ is a (γ, α)-unique-neighbor expander if for every $S \subseteq U$ such that $|S| \leq \gamma |L|$, it holds that $|\Gamma_u(S)| \geq \alpha d |S|$. The following theorem readily follows by the construction of lossless conductors as given by Theorem 7.3 in [8].

Theorem 6 ([8]). There exist universal constants $c_0 \geq 1$ and $\beta \leq 1$ such that the following holds. For every n and $m \leq n$, there exists an explicit (γ, α)-unique-neighbor expander $G = (L, R, E)$ with $|L| = 2^n$, $|R| = 2^m$, having left degree

$$d \leq 2^{c_0 \cdot \log^3(n-m)},$$

where $\alpha = \Omega(1)$, and $\gamma = \beta \cdot \frac{2^{m-n}}{d}$.

Definition 7 (Expander codes). Let $G = (L, R, E)$ be a bipartite graph and let $\mathbb{F}$ be a field. The expander code associated with G is defined by

$$\mathrm{EC}_{\mathbb{F}}(G) = \left\{ w \in \mathbb{F}^L \mid \forall v \in R \quad \sum_{u \in \Gamma(v)} w_u = 0 \right\}.$$

We usually omit the subscript $\mathbb{F}$ when the field is clear from context. It is easy to see that the rate of $\mathrm{EC}_{\mathbb{F}}(G)$ is at least $1 - \frac{|R|}{|L|}$.

Definition 8 (VLTCs). A code $C \subseteq \mathbb{F}^N$ is called a $(q, \delta, \kappa, \sigma)$-VLTC (vicinity locally testable code, abbreviated) if there exists a randomized procedure

$$\mathrm{Tes} : \mathbb{F}^N \to \{\circ, \perp\}$$

with the following guarantees:

- For every $c \in C$ and $w \in \mathbb{F}^N$, satisfying $\mathrm{RelDist}(w, c) \leq \delta$,

$$\Pr[\mathrm{Tes}(w) = \perp] \geq \kappa \cdot \mathrm{RelDist}(w, c) - \sigma;$$

- $\mathrm{Tes}(c) = \circ$ with probability one on any $c \in C$.
- $\mathrm{Tes}(w)$ always makes at most q queries to w.

We call Tes a local tester (or tester for short). The parameter q is referred to as the query complexity.

Theorem 12. For every finite field $\mathbb{F}$, n which is a power of 2, and $\delta > 0$, there exists an explicit $(q, \delta, \frac{1}{3})$-RLCC $C \subseteq \mathbb{F}^{[n]}$ with query complexity

$$q = (\log n)^{2+o(1)},$$

and rate

$$\rho = 1 - \delta \cdot 2^{O((\log\log \frac{1}{\delta})^3)} - o(1).$$

Complete author support and main construction

We move to show that expander codes constructed from unique-neighbor expanders are VLTCs.

Lemma 9. Let $G = (L, R, E)$ be a d-left-regular (γ, α)-unique-neighbor expander with average right-degree $\bar{c}$. Then, for every $b > 1$, $\mathrm{EC}(G)$ is a $(b\bar{c}, \gamma, \alpha\bar{c}, \frac{1}{b})$-VLTC.

Proof. Define

$$R' = \{v \in R \mid \deg(v) \leq b\bar{c}\}.$$

By an averaging argument, $|R'| \geq (1 - \frac{1}{b})|R|$. The tester for $\mathrm{EC}(G)$, given oracle access to $w \in \mathbb{F}^L$, proceeds as follows:

1. Sample $v \in R'$ uniformly at random.
2. Query w on $\Gamma(v)$.
3. Output $\circ$ if $\sum_{u \in \Gamma(v)} w_u = 0$; and $\perp$ otherwise.

As the sampled vertex v is in R', the query complexity of the tester is indeed bounded above by $b\bar{c}$. Further, when $w \in \mathrm{EC}(G)$, the tester outputs $\circ$ with certainty. Consider then a word $w \in \mathbb{F}^L$ such that $\mathrm{RelDist}(w, c) \leq \gamma$ for some codeword $c \in \mathrm{EC}(G)$. Let

$$S = \{v \in L \mid w_v \neq c_v\}.$$

As $|S| \leq \gamma|L|$ we have that $|\Gamma_u(S)| \geq \alpha d|S|$. Notice that if the vertex v that is sampled in Step 1 lies in $\Gamma_u(S)$ then the tester outputs $\perp$. Therefore, the

probability of the tester to output $\perp$ is at least

$$\begin{aligned}\frac{|\Gamma_u(S)| - |R \setminus R'|}{|R|} &\geq \frac{\alpha d|S|}{|R|} - \frac{1}{b} \\ &= \alpha \cdot \frac{d|L|}{|R|} \cdot \frac{|S|}{|L|} - \frac{1}{b} \\ &= \alpha\bar{c} \cdot \mathrm{RelDist}(w, c) - \frac{1}{b},\end{aligned}$$

which concludes the proof. □

We will use the following easy claim.

Claim 10. Let $C \subseteq \mathbb{F}^N$ be a $(q, \delta, \kappa, \sigma)$-VLTC with a tester Tes, and further let $c \in C$ and $w \in \mathbb{F}^N$ be such that $\alpha \leq \mathrm{RelDist}(w, c) \leq \delta$. Assume that we run $\mathrm{Tes}(w)$ for g times, independently. Then, the probability that one of the simulations outputted $\perp$ is at least $1 - e^{-\beta g}$, where $\beta = \kappa\alpha - \sigma$.

Proof. The probability that a single simulation of $\mathrm{Tes}(w)$ outputs $\perp$ is at least

$$\kappa \cdot \mathrm{RelDist}(w, c) - \sigma \geq \kappa\alpha - \sigma = \beta.$$

The probability that all the simulations output $\circ$ is thus $(1 - \beta)^g \leq e^{-\beta g}$. □

Following a similar argument to the one underlying the Kumar-Mon reduction, the following proposition states that a sequence of VLTCs can be used to construct an RLCC.

Proposition 11. Let $C_1 \subseteq \mathbb{F}^{N_1}, \ldots, C_m \subseteq \mathbb{F}^{N_m}$ be codes with rates $\rho_1, \ldots, \rho_m$, respectively, such that for every $i \in [m-1]$, C_i is a $(q', \delta', \kappa', \sigma')$-VLTC, and C_m is a $(q, \delta, \kappa, \sigma)$-VLTC. Further assume that $|N_1| \leq \frac{1}{\delta'}$, $|N_m| = n$, and for every $1 < i \leq m$, $|N_i| = 2|N_{i-1}|$. Then, for every $g \in \mathbb{N}$, there exists an $((m-1)q' + gq + 1, \delta, \varepsilon)$-RLCC $C \subseteq \mathbb{F}^{[n]}$ with rate

$$\rho \geq 1 - \sum_{i=1}^{m}(1 - \rho_i),$$

where

$$\varepsilon \leq 1 - \min\left\{\frac{\kappa'\delta'}{2} - \sigma', e^{g(\sigma - \frac{\kappa\delta'}{2})}\right\}.$$

Moreover, if the codes $C_1, \ldots, C_m$ are explicit, then so is the resulting code C.

Proof. We start by describing how the code C is constructed.

The code construction. Let $P_1, \ldots, P_m$ be an arbitrary fixed sequence of partitions of $[n]$, satisfying that for every $i \in [m]$, P_i has 2^{m-i} equal-size parts denoted $\{B_1^i, \ldots, B_{2^{m-i}}^i\}$, and that for every $1 < i \leq m$, P_{i-1} is a sub-partition of P_i (that is, for every $B \in P_{i-1}$, there exists $B' \in P_i$ such that $B \subseteq B'$). For every $i \in [m]$ and $B \in P_i$ let $f_{i,B} : B \to N_i$ be an arbitrary bijection, and define

$$C_{i,B} = \{c \circ f_{i,B} \mid c \in C_i\}.$$

Finally, define the code

$$C = \{w \in \mathbb{F}^{[n]} \mid \forall i \in [m], B \in P_i : w_B \in C_{i,B}\}.$$

The moreover part of the proof readily follows. The efficiency of the corrector will be self evident as well once the corrector is presented.

Rate analysis. For every $i \in [m]$ and $B \in P_i$, the number of linear constraints required to impose so that $w_B \in C_{i,B}$ is at most $(1-\rho_i)|N_i|$. Therefore, the total number of constraints in the definition of the code C is bounded above by

$$\sum_{i=1}^{m} |P_i|(1-\rho_i)|N_i| = \sum_{i=1}^{m} n(1-\rho_i),$$

which establishes the lower bound on the rate of C.

The corrector. We turn to describe a corrector $\mathrm{Cor} : \mathbb{F}^{[n]} \times [n] \to \mathbb{F} \cup \{\perp\}$ for C. As for every $i \in [m]$, C_i is a VLTC, it is immediate that so is $C_{i,B}$ for every $B \in P_i$, with the same parameters as C_i. The local tester for $C_{i,B}$ that is induced in the natural way from the local tester for C_i is denoted

$$\mathrm{Tes}_{i,B} : \mathbb{F}^B \to \{\circ, \perp\}.$$

Let $w \in \mathbb{F}^{[n]}$ and $j \in [n]$. Let $r_1 = r_1(j), \ldots, r_m = r_m(j)$ be the indices of blocks within the corresponding partitions $P_1, \ldots, P_m$ such that $j \in B^1_{r_1} \subseteq \cdots \subseteq B^m_{r_m}$. The corrector $\mathrm{Cor}(w, j)$ proceeds as follows:

1. For $i = 1, \ldots, m-1$, simulate $\mathrm{Tes}_{i,B^i_{r_i}}(w_{B^i_{r_i}})$.
2. Simulate $\mathrm{Tes}_{m,B^m_{r_m}}(w_{B^m_{r_m}})$ for g times.
3. If any of the simulations outputted $\perp$, output $\perp$; otherwise, output w_j.

Query analysis. As Cor simulates $m-1$ testers with query complexity q', and invokes g simulations of one tester with query complexity q, the overall query complexity is $(m-1)q' + gq + 1$, accounting also for querying w_j.

Correctness. Clearly, if w is a codeword of C then $w_{B^i_{r_i}} \in C_{i,B^i_{r_i}}$ for every $i \in [m]$, and so $\mathrm{Tes}_{i,B^i_{r_i}}(w_{B^i_{r_i}}) = \circ$ with certainty. Therefore, $\mathrm{Cor}(w, j) = w_j$ with certainty, as required. Assume that $w \in \mathbb{F}^{[n]}$ is such that $\mathrm{Dist}(w, c) \leq \delta n$ for $c \in C$. Since $\mathrm{Cor}(w, j)$ always either outputs $\perp$ or w_j, it suffices to show that if $w_j \neq c_j$ then the corrector outputs $\perp$ with probability at least $1 - \varepsilon$. Towards this end, assume $w_j \neq c_j$, and hence $w_{B^1_{r_1}} \neq c_{B^1_{r_1}}$, and further note that $w_{B^m_{r_m}} = w$ and $c_{B^m_{r_m}} = c$. For every $i \in [m-1]$ define $\delta_i = \delta'$, and let $\delta_m = \delta$. Since, per our assumption, $|N_1| \leq \frac{1}{\delta'}$, we have that

$$\mathrm{RelDist}(w_{B^1_{r_1}}, c_{B^1_{r_1}}) \geq \delta' = \delta_1,$$

whereas

$$\mathrm{RelDist}(w_{B^m_{r_m}}, c_{B^m_{r_m}}) \leq \delta = \delta_m.$$

Let $\iota \in \{2, 3, \ldots, m\}$ be any index satisfying that

$$\mathrm{RelDist}(w_{B^{\iota-1}_{r_{\iota-1}}}, c_{B^{\iota-1}_{r_{\iota-1}}}) \geq \delta_{\iota-1}$$
$$\mathrm{RelDist}(w_{B^{\iota}_{r_{\iota}}}, c_{B^{\iota}_{r_{\iota}}}) \leq \delta_{\iota}.$$

By the above account, ι is well-defined. Since $B^{\iota-1}_{r_{\iota-1}} \subseteq B^{\iota}_{r_{\iota}}$ and $|B^{\iota-1}_{r_{\iota-1}}| = \frac{1}{2}|B^{\iota}_{r_{\iota}}|$, we have that

$$\frac{\delta_{\iota-1}}{2} \leq \mathrm{RelDist}(w_{B^{\iota}_{r_{\iota}}}, c_{B^{\iota}_{r_{\iota}}}) \leq \delta_{\iota}.$$

If $\iota < m$, as $\mathrm{Tes}_{\iota, B^{\iota}_{r_{\iota}}}$ is a local tester for the $(q', \delta_{\iota}, \kappa', \sigma')$-VLTC $C_{i, B^{\iota}_{r_{\iota}}}$ and since $c_{B^{\iota}_{r_{\iota}}} \in C_{\iota, B^{\iota}_{r_{\iota}}}$, it holds that $\mathrm{Tes}_{\iota, B^{\iota}_{r_{\iota}}}(w_{B^{\iota}_{r_{\iota}}})$ outputs $\perp$ with probability at least

$$\frac{\kappa' \delta_{\iota-1}}{2} - \sigma' = \frac{\kappa' \delta'}{2} - \sigma'.$$

If otherwise $\iota = m$ then we set

$$\beta = \frac{\kappa \delta_{m-1}}{2} - \sigma = \frac{\kappa \delta'}{2} - \sigma$$

and then by Claim 10 one of the g simulations of $\mathrm{Tes}_{m, B^{m}_{r_m}}((w_{B^{m}_{r_m}}))$ with probability at least $1 - e^{-\beta g}$. Thus, $\mathrm{Cor}(w, j)$ outputs $\perp$ with probability at least $\min\{\frac{\kappa' \delta'}{2} - \sigma', 1 - e^{-\beta g}\}$, as required. □

We are now ready to prove our main theorem.

Proof of Theorem 12. Write $n = 2^r$, and let

$$\ell = r \log r = \log n \cdot \log \log n.$$

Due to the claimed tradeoff between the rate and the correction radius, we may as well assume that $\delta \geq \frac{1}{2^{\lceil \log \ell \rceil}}$. We proceed to describe the sequence of expander codes that we will use, which consists of $s = r - \lceil \log \ell \rceil + 1$ codes. For every $i \in [s]$, the block-length of the i-th code is

$$n_i = 2^{\lceil \log \ell \rceil + i - 1}.$$

Note that, in particular, $n_s = n$. Further, the number of linear constraints defining the i-th code is $m_i = 2^{i-1}$ for $i \in [s-1]$, whereas the for the s-th code,

$$m_s = 2^{r - \lfloor \log \frac{1}{\delta} + \log \beta - c_0 \log^3(\log \frac{1}{\delta}) \rfloor},$$

where β is the constant from Theorem 6. Invoking Theorem 6, for every $i \in [s]$ let $G_i = (L_i, R_i, E_i)$ be a d_i-left-regular bipartite graph with $|L_i| = n_i$ and $|R_i| = m_i$ which is a (δ_i, α)-unique-neighbor expander for $\alpha = \Omega(1)$, such that for $i \in [s-1]$,

$$d_i \leq 2^{c_0 \log^3(\lceil \log \ell \rceil)} \triangleq d,$$
$$\delta_i \geq \frac{\beta}{d \cdot 2^{\lceil \log \ell \rceil}} \triangleq \delta',$$

and

$$d_s \leq 2^{c_0 \log^3(\log \frac{1}{\delta})},$$
$$\delta_s \geq \frac{\beta}{d_s \cdot 2^{\lfloor \log(1/\delta)+\log(\beta)-c_0 \log^3(\log(1/\delta))\rfloor}} \geq \delta.$$

The sequence of codes is defined by setting, for every $i \in [s]$, $C_i = \mathrm{EC}(G_i)$. We turn to address the VLTC-ness of $C_1, \ldots, C_s$. Set $b = \frac{4}{\alpha\beta}$ and $b_s = \frac{4m_s}{\alpha\delta' d_s n_s}$. By Lemma 9, for every $i \in [s-1]$, C_i is a

$$\left(\frac{bd_i n_i}{m_i} \leq bd2^{\lceil \log \ell \rceil}, \delta', \alpha d 2^{\lceil \log \ell \rceil}, \frac{1}{b}\right)\text{-VLTC},$$

and C_s is a

$$\left(\frac{b_s d_s n_s}{m_s}, \delta, \alpha \frac{d_s n_s}{m_s}, \frac{1}{b_s}\right)\text{-VLTC}.$$

We further set $g = \frac{4m_s}{\alpha\delta' d_s n_s}$. We can now invoke Proposition 11 (indeed, the proposition's prerequisites are met, i.e., $n_1 = 2^{\lceil \log \ell \rceil} < \frac{1}{\delta'}$, $n_i = 2n_{i-1}$, and $n_s = n$) with our choice of g to obtain a code $C \subseteq \mathbb{F}^{[n]}$ which is an

$$\left((s-1)bd2^{\lceil \log \ell \rceil} + g\frac{b_s d_s n_s}{m_s} + 1 = O(sbd\ell + \delta/(\delta')^2), \delta, \varepsilon\right)\text{-RLCC},$$

where

$$\varepsilon \leq 1 - \min\left\{\frac{1}{2}\alpha d 2^{\lceil \log \ell \rceil}\delta' - \frac{1}{b}, (1/e)^{g(\alpha d_s n_s \delta'/(2m_s) - 1/b_s)} = (1/e)^{g\alpha d_s n_s \delta'/(4m_s)} = 1/e\right\}.$$

As

$$\begin{aligned}\frac{1}{2}\alpha d 2^{\lceil \log \ell \rceil}\delta' - \frac{1}{b} &= \frac{\alpha\beta}{2} - \frac{1}{b} \\ &= \frac{\alpha\beta}{2} - \frac{\alpha\beta}{4} \\ &= \frac{\alpha\beta}{4},\end{aligned}$$

we see that $\varepsilon \leq 1 - \min\{\frac{\alpha\beta}{4}, 1/e\}$. To decrease the error to $\frac{1}{3}$, we apply Claim 4 with $h = O(1)$, and get that C is also a $(q, \delta, \frac{1}{3})$-RLCC for

$$q = O(sbd\ell + \delta/(\delta')^2) = O(sd^2\ell + \delta d^2 \ell^2).$$

Recall that $s \leq \log n$,

$$d = 2^{O((\log\log r)^3)} = 2^{O((\log\log\log n)^3)}$$

and $\ell = O(\log n \cdot \log\log n)$. Therefore,

$$q = O(sd^2\ell + d^2\ell^2) = O(d^2\ell^2) = \log^2 n \cdot 2^{O((\log\log\log n)^3)} = (\log n)^{2+o(1)}.$$

Lastly, as the rate ρ_i of every code C_i in the sequence is at least $1 - \frac{m_i}{n_i}$, Proposition 11 implies that the rate ρ of C is lower bounded by

$$\begin{aligned}
\rho &\geq 1 - \sum_{i=1}^{s} \left(\frac{m_i}{n_i} \right) \\
&= 1 - (s-1)\frac{1}{2^{\lceil \log \ell \rceil}} - \frac{1}{2^{\lfloor \log \frac{1}{\delta} + \log \beta - c_0 \log^3(\log \frac{1}{\delta}) \rfloor}} \\
&= 1 - O\left(\frac{s}{\ell}\right) - \delta \cdot 2^{O(\log^3(\log \frac{1}{\delta}))} \\
&= 1 - O\left(\frac{r}{\ell}\right) - \delta \cdot 2^{O(\log^3(\log \frac{1}{\delta}))} \\
&= 1 - \delta \cdot 2^{O(\log^3(\log \frac{1}{\delta}))} - o(1).
\end{aligned}$$

This concludes the proof. □

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